HOMEWORK 9.4 · UNIT 9 · NC.8.G.9

Unit 9 Review

Targets: G.9-2, G.9-3, G.9-5, G.9-4, G.9-6, G.9-8, G.9-7, G.9-9, G.9-10

Name _____ Date _____ Block _____

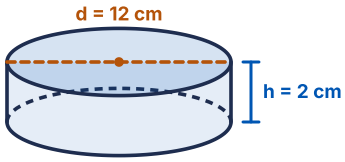
Remember: Cylinder $V = \pi r^2 h$ · Cone $V = \frac{1}{3}\pi r^2 h$ · Sphere $V = \frac{4}{3}\pi r^3$. Hemisphere = $\frac{1}{2}$ of a sphere. $r = d \div 2$. Answer in terms of π, or use 3.14 and round. Always use cubic units. A cone = $\frac{1}{3}$ of its cylinder. A sphere = $\frac{2}{3}$ of the cylinder around it.

SCORE 3.0 Approaching · the basics

Targets: G.9-2, G.9-3, G.9-5, G.9-4, G.9-6

1 REVIEW 9.2 G.9-2

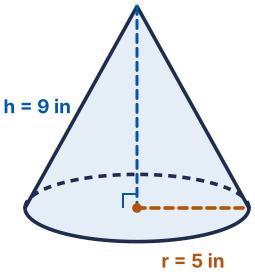
Find the volume. Answer in terms of π.



(A) 72π cm^3
(B) 288π cm^3
(C) 24π cm^3
(D) 12π cm^3

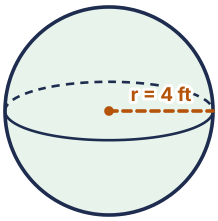
2 REVIEW 9.1 G.9-3,G.9-5

Find the volume of the cone. Use 3.14 for π. Round to the nearest tenth.



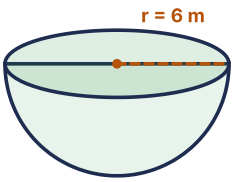
3 REVIEW 9.1 G.9-4

Find the volume of the sphere. Answer in terms of π.



4 REVIEW 9.3 G.9-4,G.9-5

Find the volume of the hemisphere. Use 3.14 for π. Round to the nearest tenth.



5 REVIEW 9.1 G.9-6

A cone holds one-third as much as a cylinder with the same base and height.

True False

SCORE 4.0 Proficient · on target

Targets: G.9-8, G.9-7

6 A cylinder holds 245π cm^3. Its height is 5 cm. Find its radius.

REVIEW 9.2 G.9-8

7 **Dev** made a mistake. Circle it. Explain it. Fix it.

REVIEW 9.2 G.9-7

DEV'S WORK

Sphere: $r = 3$ in

$V = \frac{4}{3}\pi(3)^2$

$V = 12\pi$ in^3

8 A traffic cone has a diameter of 10 in and a height of 18 in. Find its volume. Use 3.14 for π. Round to the nearest tenth.

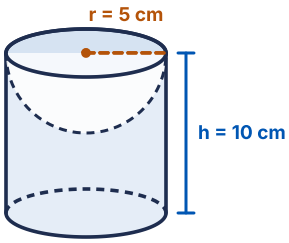
REVIEW 9.2 G.9-7

SCORE 5.0 Mastery · go deeper

Targets: G.9-9, G.9-10

9 A block is a cylinder (r = 5 cm, h = 10 cm). A hemisphere with r = 5 cm is scooped out of the top. How much is left? Answer in terms of π. Then round.

REVIEW 9.3 G.9-9



Cylinder = _____ . Hemisphere = _____ . Left = _____ .

10 Cone A has r = 2 and h = 3. Cone B triples the height. Cone C triples the radius. How many times more does each hold? Explain.

REVIEW 9.3 G.9-10

B holds _____ times. C holds _____ times, because _____ .